

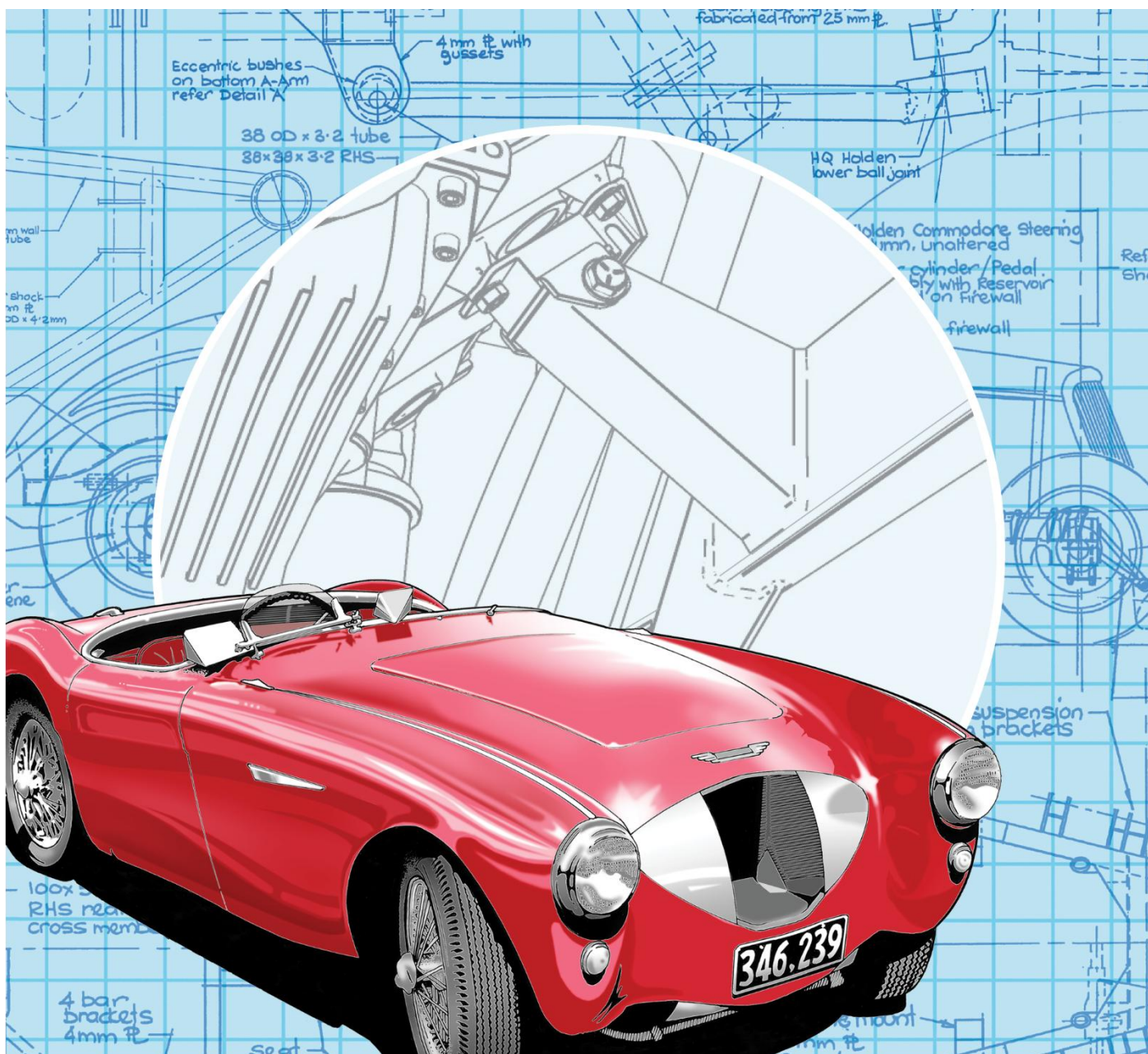
Helping New Zealanders Build & Modify Safe Vehicles

Low Volume Vehicle Standard

85-40(05)

Engine & Drive-train

Version 6 | Effective from 1 September 2026



Approval Record

Signed in accordance with clause 1.3(5) of the <i>Low Volume Vehicle Code</i> of LVVTA, on by:			
New Zealand Transport Agency		Low Volume Vehicle Technical Association	
Name	Signature	Name	Signature
	_____		_____

Amendment Record

Amendments details	Version #	Issue date	Effect date
• LVV Standard 85-40(00) - Original Issue	Version 1	1 September 2002	1 September 2002
• LVV Standard 85-40(01)	Version 2	1 July 2016	1 July 2016
• LVV Standard 85-40(02)	Version 3	25 October 2016	25 October 2016
• LVV Standard 85-40(03)	Version 4	5 February 2021	1 March 2021
• LVV Standard 85-40(04)	Version 5	1 June 2021	1 July 2021
• LVV Standard 85-40(05)	Version 6	1 August 2026	1 September 2026
Note 1 Text highlighted in grey shows amendments that have been made subsequent to the previous version of this standard, and a grey vertical stroke to the left of the text denotes important new or changed information (which may include information which has been removed).			

About Low Volume Vehicle Standards

The Low Volume Vehicle Standards (the standards) specify that vehicles (which have been modified or constructed) require low volume vehicle certification, and provide a set of overarching technical requirements that the vehicles must meet. The standard refers to the associated *New Zealand Car Construction Manual chapter*, that sets out the specific technical requirements which must be met in order to achieve compliance with the standard.

Author, Publisher, & Owner

This standard is authored, published, and owned by the Low Volume Vehicle Technical Association Incorporated (LVVTA). LVVTA is an incorporated society established in 1992, that represents a group of specialist automotive organisations (in turn representing approximately 150,000 members) who are dedicated to ensuring that vehicles, when scratch-built or modified, meet the highest practicable safety standards.

The information in this standard has stemmed from work undertaken by LVVTA founding member organisations that commenced in 1989 and has been progressively developed as an integral part of the New Zealand Government's land transport regulatory system, by agreement and in consultation with the New Zealand Transport Agency (NZTA).

As a result, the considerable experience in applied safety engineering built up by LVVTA and its specialist automotive member groups over the past several decades can be of benefit to members of the New Zealand public who also wish to build or modify motor vehicles.

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Availability and Current Version

This standard is printed and distributed by LVVTA, and is available to the public free of charge from the LVVTA website; www.lvvta.org.nz.

Note that printed copies of this standard, like any other printed LVVTA documents, may have been superseded by a later version and become out of date.

Therefore, this and all other LVVTA documents should not be relied upon without first ensuring that the version number (on the right-hand side of the header above) is the current version – please visit the LVV Standards area of www.lvvta.org.nz to check that this standard is in fact the latest version.

Users' Feedback

This standard is constantly undergoing an evolutionary development process in order to keep pace with changing trends and technology. To assist in this, LVVTA invites users of this standard to engage in an ongoing consultation process with us by making submissions for any changes, additions, or clarifications which might improve the standard, at any time.

Any submissions made via this rolling consultation process will be thoroughly considered, and incorporated, where appropriate, at the next available amendment opportunity.

Submissions should be made to submission@lvvta.org.nz, with the name of this standard in the Subject line.

Supporting Information

This standard should be read in conjunction with *Chapter 9: Engine & Drive-train* of the *New Zealand Car Construction Manual*, which provides all necessary requirements and best-practice guidance to enable compliance with this standard.

Legal Status & Copyright

This standard is incorporated by reference, as an integral part of the *Low Volume Vehicle Code*, within *Land Transport Rule: Vehicle Standards Compliance 2002*. This standard corresponds technically with *Land Transport Rule 32014: Light-vehicle Brakes 2002*, and *Land Transport Rule 32003/1 Steering Systems 2001*.

The content of this standard remains the property of LVVTA, and no part may be reproduced without LVVTA's written consent.

Disclaimer

LVVTA has made all reasonable efforts to provide sound and correct advice, based on the historical knowledge and best practice experiences of all parties involved in the development and production of this standard.

However, no responsibility or liability is accepted by LVVTA for any error or omission, or any loss suffered by any person relying directly or indirectly on this standard. Any person who builds or modifies a motor vehicle accepts that there may be some associated risks, and does so in the full knowledge of this, and accepts full responsibility for their own actions.

Credits

LVVTA acknowledges the following contributors for their assistance in the development of this standard:

- Cover blueprint drawings and cover central CAD diagram: Graeme Walls
- Cover car illustration: Tony Johnson

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85-40(05)

Engine & Drive-train

Purpose of this Standard

The purpose of this low volume vehicle standard (the standard) is to provide the legal framework for the low volume vehicle certification of a vehicle which has been scratch-built, or has undergone an engine or drive-train modification or conversion which may have affected the vehicle's braking or steering system, general safe operation, or durability

Such a vehicle is required by the *Land Transport Rule 32014: Light-vehicle Brakes 2002* to comply with the *Low Volume Vehicle Code*, which in turn, requires compliance with this standard.

This standard sets out:

- which vehicles are required to meet this standard; and
- how this standard is applied to those vehicles; and
- the technical requirements which must be met by vehicles required to comply with this standard.

Section 1 Scope and Application of this Standard

1.1 Scope of this standard

1.1(1) This standard applies to all light vehicles other than those specified in section 3, that (see Note 1 below):

(a) are modified either:

- (i) on or after 1 January 1992 in such a way that the vehicle's safe operation may, directly or indirectly, be affected as a result of an engine or drive-train conversion or modification; or
- (ii) before 1 January 1992 in such a way that the vehicle's safe operation may, directly or indirectly, be affected as a result of an engine or drive-train conversion or modification, and have not been continuously registered in New Zealand since that date;

or

(b) are scratch-built either:

- (i) on or after 1 January 1992; or
- (ii) before 1 January 1992 and have not been continuously registered in New Zealand since that date.

Note 1 For the avoidance of doubt, 1.1(1) specifies that a low volume vehicle (which has been scratch-built, or modified in such a way that its safe operation may, directly or indirectly, be affected as a result of an engine or drive-train conversion or modification) is not required to be certified to this standard only if the vehicle:

- was scratch-built or modified prior to 1 January 1992; and
- has been issued with a valid *Modification Declaration Certificate*; and
- has been continuously registered in New Zealand since 1 January 1992; and
- has not been subsequently modified.

1.2 Application of this standard

- 1.2(1) A light vehicle that is modified or scratch-built as in 1.1(1), becomes a low volume vehicle, and must either (see Notes 1 and 2 below):
- (a) in the case of a vehicle that was scratch-built or modified before 1 January 1992 and has not been continuously registered in New Zealand, or a vehicle that was scratch-built or modified between 1 January 1992 and 1 September 2002:
 - (i) be certified in accordance with the procedures specified in *chapter 2* of the *Low Volume Vehicle Code*; and
 - (ii) comply with the general safety requirements in 2.1; and
 - (iii) comply with those technical requirements in 2.2 and 2.3, as determined to be appropriate by a Low Volume Vehicle Certifier;
 - or
 - (b) in the case of a vehicle that was scratch-built or modified after 1 September 2002:
 - (i) be certified in accordance with the procedures specified in *chapter 2* of the *Low Volume Vehicle Code*; and
 - (ii) comply with the general safety requirements in 2.1; and
 - (iii) comply with all applicable technical requirements in 2.2 and 2.3.

Note 1 The date of 1 September 2002 specified in 1.2(1)(a) and 1.2(1)(b) is the implementation date of LVV Standard 85-40: Engine & Drive-train (Version 1), that introduced specific technical requirements which must be met.

Note 2 In the case of a vehicle specified in 1.2(1)(a), an LVV Certifier must apply the applicable technical requirements referred to in 2.2 and 2.3 as a guideline upon which to base their discretionary judgement on the safety of the vehicle, taking into account the modification or construction date of the vehicle.

Section 2 Technical Requirements of this Standard

2.1 General safety requirements

- 2.1(1) A low volume vehicle must (see Note 1 below):
- (a) be designed and constructed using materials and components that are fit for their purpose; and
 - (b) be safe to be operated on the road.

Note 1 The requirements specified in 2.1(1) are selected from 2.3 of *Part 2* of the *Low Volume Vehicle Code*, reproduced here in the interest of convenience, and are over-riding requirements which make it clear that, regardless of what technical requirements are or are not in place, every vehicle certified to the *Low Volume Vehicle Code* must be fit for its purpose, and must be safe.

- 2.1(2) A low volume vehicle must meet all applicable general safety requirements contained in (see Notes 1 and 2 below):
- (a) *Land Transport Rule: Steering Systems 2001*; and
 - (b) *Land Transport Rule: Light-vehicle Brakes 2002*; and

(c) *Land Transport Rule: Vehicle Equipment 2004.*

Note 1	Every vehicle must comply with all general safety requirements listed in all applicable <i>Land Transport Rules</i> . The <i>Land Transport Rules</i> listed above are those <i>Rules</i> which correspond technically with this low volume vehicle standard, and therefore contain general safety requirements with which a low volume vehicle certified to this standard must comply.
Note 2	The applicable general safety requirements from the <i>Land Transport Rules</i> listed above are either: • incorporated within those <i>New Zealand Car Construction Manual Chapters</i> referenced in 2.2; or • incorporated within other <i>New Zealand Car Construction Manual Chapters</i> which the <i>chapters</i> referenced in 2.2 require compliance with.

2.2 Overarching technical requirements

- 2.2(1) An engine-mounting system in a low volume vehicle must be of good design and construction, must position and support the engine correctly, and must be securely attached.
- 2.2(2) All ancillary systems potentially affected by an engine conversion or modification in a low volume vehicle, including accelerator, electrical, fuel, vacuum, oiling, braking, steering, or exhaust, must not cause the vehicle to be unsafe.
- 2.2(3) An engine conversion or modification in a low volume vehicle must not cause the driver's vision to be impaired.
- 2.2(4) A gearbox-mounting system in a low volume vehicle must be of good design and construction, must position and support the gearbox correctly, and must be securely attached.
- 2.2(5) A gearbox conversion in a low volume vehicle, including gear selection and inhibitor switch operation, must not cause the vehicle to be unsafe.
- 2.2(6) A drive-shaft in a low volume vehicle must be of good design and appropriate strength, must be fitted with suitable drive-shaft universals that operate correctly and safely, and must be securely attached.
- 2.2(7) A drive-shaft in a low volume vehicle must, where applicable, be safely contained by a drive-shaft safety-loop which is of good design and appropriate strength, and be securely attached.
- 2.2(8) An axle housing in a low volume vehicle must be securely attached, and must not incorporate any modifications that may cause the low volume vehicle to which it is fitted to be unsafe.
- 2.2(9) A change to a driving-wheel configuration in a low volume vehicle must not cause the vehicle to be unsafe.
- 2.2(10) In order to demonstrate compliance with 2.2(1) to 2.2(9), a low volume vehicle must comply with all applicable technical requirements specified in *Chapter 9: Engine & Drive-train* of the *NZ Car Construction Manual*; and

Note 1	For the avoidance of duplication, the technical requirements for engine and drive-train conversions and modifications are contained in <i>Chapter 9: Engine & Drive-train</i> of the <i>New Zealand Car Construction Manual</i> .
Note 2	For the avoidance of doubt, a low volume vehicle to which 1.2(1)(a) applies, must comply with only the relevant requirements specified in <i>Chapter 9: Engine & Drive-train</i> of the <i>New Zealand Car Construction Manual</i> considered appropriate by an LVV Certifier, taking into account the modification or construction date of the vehicle.
Note 3	An electronic version of <i>Chapter 9: Engine & Drive-train</i> , and all other chapters of the <i>New Zealand Car Construction Manual</i> , can be accessed from www.lvvt.org.nz either individually or as a complete Manual, free of charge.
Note 4	Printed copies of LVV documents may become out of date and should not be relied upon without ensuring that the version is current – visit www.lvvt.org.nz to check that any document referred to is the latest version.

2.3 Associated Low Volume Vehicle Standards

- 2.3(1) A low volume vehicle that is required to comply with this standard must also comply with the applicable requirements of (see Notes 1 and 2 below):
- (a) Low Volume Vehicle Standard 35-00: Braking Systems; and
 - (b) Low Volume Vehicle Standard 195-00: Suspension Systems; and
 - (c) Low Volume Vehicle Standard 90-10: Exhaust Gas Emissions; and
 - (d) Low Volume Vehicle Standard 90-20: Exhaust Noise Emissions.
- 2.3(2) A low volume vehicle that has been converted to electric or hybrid power and is required to comply with this standard, must also comply with Low Volume Vehicle Standard 75-00 (Electric & Hybrid Vehicles).

Note 1 All standards referred to in 2.3(1) and 2.3(2) can be accessed from www.lvvta.org.nz and are free of charge.

Note 2 Printed copies of LVV documents may become out of date and should not be relied upon without ensuring that the version is current – visit www.lvvta.org.nz to check that any document referred to is the latest version.

Section 3 Vehicles Not Required to Comply With this Standard

3.1 Specified vehicle classes

- 3.1(1) A vehicle is not required to comply with this standard if the vehicle is:
- (a) a powered bicycle of Class AB; or
 - (b) a motorcycle or moped of Class LA, LB, LC, LD, or LE; or
 - (c) a light trailer of Class TA or TB.

3.2 Enforcement, emergency, & second-stage manufactured vehicles

- 3.2(1) A vehicle is not required to comply with this standard if the vehicle is either:
- (a) modified for the purpose of law enforcement or provision of emergency services; or
 - (b) identified as having been modified by a second-stage vehicle manufacturer, and complies with an approved overseas standard that is listed in Annex 6 of the Low Volume Vehicle Code, and has not been subsequently modified further.

3.3 Specified modifications

- 3.3(1) A modification to a vehicle is not required to comply with this standard if the modification:
- (a) has not compromised the safe performance of the vehicle; and
 - (b) is listed as a modification which does not require low volume vehicle certification, within either:
 - (i) the LVVTA Modification Threshold Schedule; or
 - (ii) a relevant Modification Table in the Vehicle Inspection Requirements Manual of the New Zealand Transport Agency